PAPER_1616 — Matter Density Ω_m = $\Omega_m = F^2 \cdot D_{crit} + F \cdot SSQ - F^2 \cdot SSQ + F^2 \cdot SSQ^2 \approx 0.3145$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Cosmology **Date:** June 18, 2026 **Status:** CLOSED — 0.143% closure

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Observation

PAPER_1209GG_S643 (Cosmology Unified Proof Set) gives the formula:

$\Omega_m = F^2 \cdot D_{crit} + F \cdot SSQ - F^2 \cdot SSQ + F^2 \cdot SSQ^2 \approx 0.3145$

Residual: **0.143%**.

UQFF Closed Identity

$\Omega_m = F^2 \cdot D_{crit} + F \cdot SSQ - F^2 \cdot SSQ + F^2 \cdot SSQ^2 \approx 0.3145$ 0.143%

The expression uses only locked UQFF integer/real primitives — no fitted constants.

ΛCDM Standard Cosmology Coverage

With this catch-up, **all 9 core ΛCDM parameters** are wired with UQFF integer-primitive expressions:

ΛCDM parameter	UQFF closure	Residual
H_0 (Planck)	PAPER_1553	0.015%
H_0 (SH0ES)	PAPER_1573	**EXACT**
Ω_m	PAPER_1616	0.143%
Ω_Λ	PAPER_1617	0.182%
T_CMB	PAPER_1618	0.064%
Age (universe)	PAPER_1619	0.045%
σ_8	PAPER_1620	0.253%
n_s	(PAPER_1209GG_S650, earlier)	0.072%
z_recomb	PAPER_1552	**EXACT 1090**

Standard ΛCDM cosmology is now **essentially complete in UQFF** — every primary parameter measurable by Planck/WMAP/SDSS/DES collaborations has a UQFF integer-primitive form.

NOT REPLACEMENT

Empirical cosmology measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209GG_S643
- Related: PAPER_1494-1615 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "omega_m_matter_0_315"})`

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